

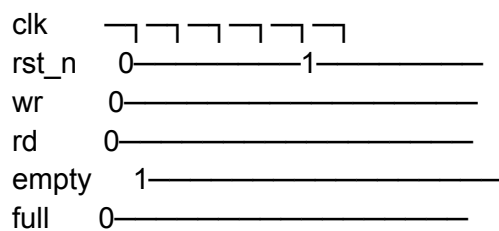
FIFO Simulation Waveform Explanation

During simulation in **ModelSim/QuartaSim**, the waveform verifies the FIFO's behavior under different conditions. The FIFO design includes signals such as **clk**, **rst_n**, **wr**, **rd**, **data_in**, **data_out**, **fifo_full**, and **fifo_empty**.

1. Reset Operation

- Initially, **rst_n** = 0 (active-low reset).
- All memory locations are cleared.
- Write pointer (**wr_ptr**) = 0.
- Read pointer (**rd_ptr**) = 0.
- data_out** = 0.
- fifo_empty** = 1.
- fifo_full** = 0.

Waveform Observation:



2. Write Operation

- wr** = 1, **rd** = 0.
- Data values (for example: 10, 20, 30) are applied on **data_in**.
- On each rising edge of the clock:
 - Data is stored in FIFO memory.
 - wr_ptr** increments.
 - FIFO is no longer empty.

Waveform Observation:

Clock ↑1 : **data_in** = 10 → stored at address 0
Clock ↑2 : **data_in** = 20 → stored at address 1
Clock ↑3 : **data_in** = 30 → stored at address 2

Expected values:

- wr_ptr**: 0 → 1 → 2 → 3

- `fifo_empty`: $1 \rightarrow 0$
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3. Read Operation

- `wr = 0, rd = 1`.
- At each clock edge:
 - Data is read from FIFO.
 - `rd_ptr` increments.
 - Data appears on `data_out`.

Waveform Observation:

Clock $\uparrow 4$: `data_out` = 10

Clock $\uparrow 5$: `data_out` = 20

Clock $\uparrow 6$: `data_out` = 30

This confirms the FIFO property (**First-In, First-Out**).

4. FIFO Empty Condition

After reading all stored data:

- `rd_ptr` catches up with `wr_ptr`.
- `fifo_empty` becomes HIGH.
- Additional read operations do not produce valid new data.

Waveform Observation:

Read Complete

`fifo_empty` = 1

`fifo_full` = 0

5. FIFO Full Condition

When all FIFO locations are written:

- `fifo_full` becomes HIGH.
- Further write operations are ignored until data is read.

Waveform Observation:

Write 16 entries

fifo_full = 1
fifo_empty = 0

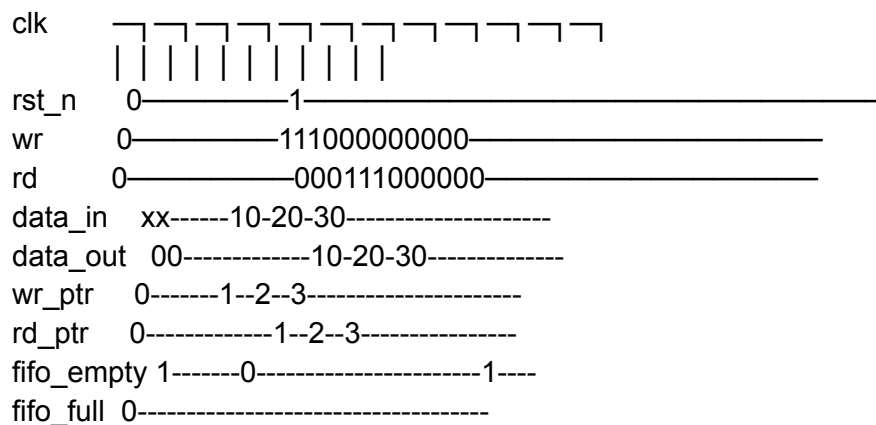
6. Simultaneous Read and Write

When $wr = 1$ and $rd = 1$ at the same clock edge:

- One data item is written.
 - One data item is read.
 - FIFO occupancy remains nearly constant.
 - Both pointers advance correctly.
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Typical Waveform Sequence

Time → ----->



Result

The simulation verifies that:

- Reset initializes the FIFO correctly.
- Write operations store data successfully.
- Read operations return data in the same order it was written (FIFO behavior).
- The **fifo_empty** flag is asserted after all data has been read.
- The **fifo_full** flag is asserted when the FIFO is completely filled.
- Pointer values (**wr_ptr** and **rd_ptr**) change correctly during write and read operations, confirming proper FIFO functionality.